CEEBB
EEDBA
DCDCE
ABAAC
BCBAB
CCBEA
BCCBC
D

5.3 perfect competitive factor market

1.

Number of Workers	Hourly Wage	Marginal Factor Cost	Marginal Revenue Product
10	$5.00		
11	5.10	$6.10/hr	$8.70/hr
12	5.20	6.30/hr	7.60/hr
13	5.30	6.50/hr	6.50/hr
14	5.40	6.70/hr	5.40/hr
15	5.50	6.90/hr	4.30/hr

According to the information in the table above, the twelfth worker would increase the hourly profit by

(A) $0.20
(B) $1.10
(C) $1.30 ✓
(D) $2.40
(E) $5.20

2. ABC Limited, Inc., sells its product in a perfectly competitive market for a price of $15 per unit and hires workers at a daily wage of $75. Labor is the only factor cost, and the firm is currently earning profits. If ABC hires one more worker and output increases by 5 units per day, the firm's profits will

(A) decrease by $5
(B) decrease by $75
(C) increase by $75
(D) increase by $15
(E) remain unchanged ✓

3. A contractor is employing labor and capital to build an office complex. At the current mix of inputs, the marginal product of labor is 30 square feet per day, and the marginal product of capital is 90 square feet per day. The price of labor is $1,000 per day, and the price of capital is $3,000 per day. To hire inputs in a cost-minimizing way, the firm should

(A) hire more labor and less capital
(B) hire less labor and more capital
(C) hire more labor and more capital
(D) hire less labor and less capital
(E) make no changes to the mix of inputs ✓

4. The marginal revenue product of labor is the

5.3 perfect competitive factor market

(A) product price times the wage rate

(B) additional revenue a firm earns when it employs an additional unit of labor ✓

(C) increase in the average product of labor when the firm employs an additional unit of labor

(D) increase in the price of labor when the firm employs an additional unit of labor

(E) marginal revenue plus product price

A firm is currently employing a cost-minimizing combination of labor and capital for a given level of output. The firm is employing 20 workers, and the marginal product of the last worker is 40 units of output. The firm is employing 100 units of capital, and the marginal product of the last unit of capital is 10 units of output.

5. If the wage rate is $20 per hour, what is the price of capital?

(A) $2.5 per hour

(B) $5 per hour ✓

(C) $10 per hour

(D) $100 per hour

(E) $200 per hour

---

6. A firm produces truffles by using labor and capital. The price of labor is $10 per unit, and the price of capital is $20 per unit. At current output level, the marginal product of labor is 40 truffles and the marginal product of capital is 60 truffles. To reduce the total cost of producing the current quantity of truffles, how should the firm change its spending on labor and capital?

5.3 perfect competitive factor market

(A)

Labor	Capital
No change	Increase

(B)

Labor	Capital
Decrease	No change

(C)

Labor	Capital
Decrease	Increase

(D)

Labor	Capital
Increase	No change

(E) ✓

Labor	Capital
Increase	Decrease

7. A firm sells its output in a perfectly competitive market and hires two inputs, capital and labor, in perfectly competitive factor markets. The product price is $15 per unit, the wage is $75 per day, and the marginal product of capital is 3. If the firm is choosing the least-cost combination of labor and capital, the firm's marginal product of labor and the price of capital must be equal to which of the following?

5.3 perfect competitive factor market

(A)

Marginal Product of Labor	Price of Capital
25	$5

(B)

Marginal Product of Labor	Price of Capital
25	$45

(C)

Marginal Product of Labor	Price of Capital
5	$15

(D)

Marginal Product of Labor	Price of Capital
5	$25

(E) ✓

Marginal Product of Labor	Price of Capital
5	$45

A firm uses capital and labor in its production process. The marginal product for the last unit of labor is 5, the marginal product for the last unit of capital is 10, and the wage is $10.

8. At what cost of hiring each unit of capital would the firm be minimizing the cost of the current output?

(A) $5
(B) $10
(C) $15
(D) $20 ✓
(E) $50

5.3 perfect competitive factor market

9. A perfectly competitive producer of steel rods and steel beams employs 100 workers with identical skills. If steel rods and steel beams sell for the same price, which of the following rules should the producer always follow to use the 100 workers efficiently?

I. Allocate workers so that the average cost of producing beams equals the average cost of producing rods.

II. Allocate workers so that the marginal product of labor is the same in both rod production and beam production.

III. Allocate half the workers to rod production and half the workers to beam production.

(A) I only

(B) II only ✓

(C) III only

(D) II and III only

(E) I, II, and III

10. A perfectly competitive firm hires three workers in a perfectly competitive labor market. The marginal products of the three workers are shown in the table below.

Worker	Marginal Product
1	50
2	30
3	20

Which of the following will be true?

(A) Each worker will receive a wage based on the marginal product of the last worker hired. ✓

(B) Each worker will receive a wage based on the marginal product of the first worker hired.

(C) Each worker will receive a wage based on the average of the marginal products of the workers.

(D) Worker 1 will receive the highest wage.

(E) Worker 3 will receive the highest wage.

11. A perfectly competitive firm is currently producing at the profit-maximizing output level. If the marginal physical product of labor is 10 units per hour and the firm pays a wage rate of $8 per hour, which of the following is true?

(A) The marginal revenue product of labor is $80.

(B) The marginal cost is $1.25.

(C) The average total cost is $0.80 per unit.

(D) The output price is $0.80 per unit. ✓

(E) The output price is $1.25 per unit.

5.3 perfect competitive factor market

12. A perfectly competitive profit-maximizing firm will continue to hire additional units of an input as long as the

(A) marginal product of the input is maximized
(B) marginal cost of employing the input is maximized
(C) value of the marginal product of the input exceeds the price of the input ✓
(D) revenue associated with the employment of the last unit of the input is maximized
(E) price of the product is greater than the marginal cost of the input

13. A profit-maximizing firm should hire an input up to the point at which

(A) marginal physical product equals marginal revenue product
(B) marginal revenue equals marginal factor cost
(C) marginal revenue product equals marginal revenue
(D) marginal revenue product equals marginal factor cost ✓
(E) marginal physical product equals marginal factor cost

14. A profit-maximizing firm that sells its output in a perfectly competitive market hires two additional workers, calculating that the contribution to total revenue of the last worker hired just equals the extra cost of hiring that worker. Six months later, the firm finds that the last worker's contribution to total revenue is less than the extra cost of hiring that worker. Which of the following may have occurred in the interim to explain this change?

(A) The firm laid off some workers.
(B) The market supply for the firm's product decreased.
(C) The market demand for the firm's product decreased. ✓
(D) A technological advance increased the productivity of the firm's workforce.
(E) The firm negotiated new contracts with workers, lowering wages.

15. A profit-maximizing firm will continue to hire workers until the marginal revenue product of labor is equal to the

(A) marginal product of labor
(B) marginal revenue product of capital
(C) demand for labor
(D) price of the good that labor is producing
(E) marginal factor cost ✓

16. A profit-maximizing firm will continue to hire workers until the

(A) marginal factor cost of labor is equal to the marginal revenue product of labor ✓
(B) marginal factor cost of labor is equal to the price of the good produced using labor
(C) marginal product of labor is equal to the marginal revenue product of labor
(D) marginal product of labor is equal to the marginal factor cost of labor
(E) marginal product of labor is equal to the price of labor

5.3 perfect competitive factor market

17. A profit-maximizing firm will hire

(A) labor until its wage rate equals its average revenue product

(B) labor until its wage rate equals its marginal revenue product ✓

(C) labor until its wage rate equals the interest rate

(D) capital until the interest rate equals the wage rate

(E) capital until the interest rate exceeds the wage rate

18. Assume a firm uses only two inputs, capital (K) and labor (L), to produce its output. Let the marginal product of capital be MP_K , the marginal product of labor be MP_L , the price of capital be P_K , and the price of labor be P_L . The least-cost combination of capital and labor needed to produce a given level of output is given by which of the following?

(A) $MP_L / P_L = MP_K / P_K$ ✓

(B) $MP_L / P_L > MP_K / P_K$

(C) $MP_L / P_K = MP_K / P_L$

(D) $(MP_L)P_L = (MP_K)P_K$

(E) $MP_L = MP_K$

19. Assume a perfectly competitive labor market. Which of the following correctly describes the individual firm's demand curve for labor and the market demand curve for labor?

5.3 perfect competitive factor market

(A) ✓

Firm's Demand Curve for Labor	Market Demand Curve for Labor
Downward sloping	Downward sloping

(B)

Firm's Demand Curve for Labor	Market Demand Curve for Labor
Downward sloping	Horizontal

(C)

Firm's Demand Curve for Labor	Market Demand Curve for Labor
Horizontal	Downward Sloping

(D)

Firm's Demand Curve for Labor	Market Demand Curve for Labor
Horizontal	Horizontal

(E)

Firm's Demand Curve for Labor	Market Demand Curve for Labor
Horizontal	Upward Sloping

20. Assume labor and capital are substitute inputs. A manufacturer will employ more labor if

(A) the price of labor increases

(B) the marginal product of labor decreases

(C) the price of capital increases ✓

(D) the marginal product of capital increases

(E) the product's demand decreases

21. Assume that a firm is hiring labor in a perfectly competitive labor market. If the marginal revenue product of labor is greater than the wage rate, which of the following will be true?

5.3 perfect competitive factor market

(A) The firm must be losing money.

(B) The firm should employ more workers. ✓

(C) The firm should replace workers with capital.

(D) The firm is maximizing its profits.

(E) The firm is experiencing diminishing marginal utility.

22. Assume that a firm uses labor and capital to produce a product. The firm hires labor at a wage rate of $4 per unit and rents capital at $5 per unit. At its current output level, the marginal physical products of labor and capital are 20 and 30 units, respectively. To minimize its cost of production without changing the level of output, the firm should

(A) make no changes

(B) hire more labor and rent more capital

(C) hire less labor and rent more capital ✓

(D) hire more labor and rent less capital

(E) hire less labor and rent less capital

23. Assume that a profit-maximizing, perfectly competitive firm hires labor in a perfectly competitive labor market. If the market wage is $12 per hour and the price of the product is $3 per unit, the firm will

(A) hire more workers if each worker can produce 3 units per hour

(B) hire another worker if the output per hour of the additional worker exceeds 4 units ✓

(C) hire fewer workers, since the hourly wage exceeds the cost of producing one unit of output

(D) not hire any workers, since the cost of labor is greater than the price of the output

(E) continue to operate in the short run but not in the long run

24. Assume that both the product and labor markets are perfectly competitive. It would be profitable for a firm to hire additional labor if the ratio of the wage to the marginal product of labor is

(A) less than the output price ✓

(B) less than the marginal cost

(C) greater than the output price

(D) greater than the marginal cost

(E) equal to the output price

25. Assume that labor and capital are the only two inputs a perfectly competitive firm uses to produce wheat. The firm hires its inputs in perfectly competitive input markets. The unit price of labor is $8 and of capital is $20. When the firm employs the profit-maximizing combination of these two inputs, the marginal product of labor is 2 tons of wheat and of capital is 5 tons of wheat. The price of wheat per ton must be

5.3 perfect competitive factor market

(A) $ 2

(B) $ 4 ✓

(C) $ 5

(D) $ 8

(E) $ 10

26. Assume that the last worker a firm hired produces 60 additional units of output per hour and the last machine rented produces 6,000 units of output per hour. A worker's hourly wage rate is $12, and the rental cost of a machine is $1,000 per hour. In order to minimize the cost of its current output, the firm should

(A) do nothing, because the costs of production are minimized

(B) increase the use of labor and decrease the use of capital

(C) increase the use of capital and decrease the use of labor ✓

(D) increase the use of labor and increase the use of capital

(E) decrease the use of labor and decrease the use of capital

27.

Number of Workers	Number of Sandwiches Produced per Day
1	80
2	150
3	200
4	240
5	250
6	230
7	200

Given the production information in the table above, how many workers would be employed if the wage rate were $20.00 per day and if sandwiches sold for $0.50?

(A) 1

(B) 2

(C) 4 ✓

(D) 5

(E) 7

5.3 perfect competitive factor market

28. At its current employment level of labor and capital, a firm observes the following.

- Marginal product of labor = 30 units
- Marginal product of capital = 60 units
- Price of labor = $3 per unit
- Price of capital = $15 per unit

Which of the following actions should the firm take in order to achieve the least-cost combination of labor and capital and produce the same level of output?

(A)

Labor	Capital
Increase	Increase

(B) ✓

Labor	Capital
Increase	Decrease

(C)

Labor	Capital
Decrease	Increase

(D)

Labor	**Capital**
Decrease	Decrease

(E)

Labor	Capital
No change	No change

5.3 perfect competitive factor market

29. For a certain firm, the marginal revenue product for the last unit of labor is $60, and the marginal revenue product for the last unit of capital is $100. Which of the following combinations of factor prices would be necessary for the firm to maximize profits?

(A)

Price of Labor	Price of Capital
$2	$5

(B)

Price of Labor	Price of Capital
$3	$20

(C)

Price of Labor	Price of Capital
$10	$10

(D)

Price of Labor	Price of Capital
$2	$25

(E) ✓

Price of Labor	Price of Capital
$60	$100

A profit-maximizing firm hires labor in a perfectly competitive market. Labor is the only variable input, and the marginal product of the last worker hired is 10 units per hour.

30. If the hourly wage is $20, the firm's marginal revenue

(A) is $2 ✓

(B) is $20

(C) increases as more output is produced

(D) increases first and then decreases as more output is produced

(E) decreases first and then increases as more output is produced

5.3 perfect competitive factor market

Answer A

Correct. A firm determines the profit-maximizing quantity of labor by equating the marginal revenue product of labor (MRPL) to the marginal factor cost (MFC), which is equal to the wage rate in a perfectly competitive market. MRPL equals the marginal product of the last worker times the marginal revenue. At the profit-maximizing quantity of labor the MRPL equals the MFC, that is, $(\text{MP} \times \text{MR}) = \text{MFC}$; thus the marginal revenue must be $\$20/\$10 = \$2$.

The following questions are based on the following output and total revenue for a monopolist in the product market.

Output Quantity (units)	Total Revenue ($)
0	0
1	11
2	20
3	27
4	32
5	35
6	36
7	35

The marginal product of labor is constant at 1 unit for every additional unit of labor. The monopolist hires labor in a perfectly competitive market at the wage rate of $5 per hour.

31. What is the marginal revenue product of labor when the firm produces the third unit of output?

(A) $3

(B) $7 ✓

(C) $9

(D) $20

(E) $27

32. How many units of labor should the firm employ to maximize profits?

5.3 perfect competitive factor market

(A) 1
(B) 2
(C) 4 ✓
(D) 5
(E) 6

---

33. Marginal resource (factor) cost can be defined as

(A) the change in total resource cost caused by the production of one more unit of output
(B) the change in total revenue caused by the addition of one more unit of a resource
(C) the change in total resource cost caused by the addition of one more unit of a resource ✓
(D) the total resource cost divided by the number of inputs used
(E) total resource cost divided by output

The following questions are based on the following table below, which shows the relationship between the number of workers and coal output (in tons per day).

Number of Workers	Total Output of Coal
0	0
1	25
2	44
3	60
4	70
5	75

34. How many workers would the coal company want to hire if the price of coal were competitively priced at $5 per ton and the wage rate were $40 per day?

(A) 5
(B) 4 ✓
(C) 3
(D) 2
(E) 0

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35. Firm Z is producing 50 widgets using labor and capital. The marginal product of the last unit of labor employed is 500; the marginal product of the last unit of machinery (or capital) is 1000. The unit price of labor is $\$10$, and the unit price of machinery or capital is $\$100$. With competitive input markets, which of the following statements is true?

5.3 perfect competitive factor market

(A) The firm is minimizing the cost of producing 50 widgets with this combination of labor and capital.

(B) The marginal product of labor is 500 and the marginal product of capital is 1000, so the firm is employing too much labor and too few machines to produce the 50 widgets.

(C) The marginal product per dollar of labor exceeds the marginal product per dollar of machinery, so more labor and less capital should be used to produce the 50 widgets. ✓

(D) The firm should lower the price of machinery so that it is equal to the price of labor.

(E) The firm is producing the given quantity inefficiently and should shut down.

Answer C

Correct. The rule for minimizing cost in the long run is to equate the marginal product per dollar of each input, i.e., $\frac{\text{MP}_\text{L}}{\text{P}_\text{L}} = \frac{\text{MP}_\text{K}}{\text{P}_\text{K}}$. Thus, $\frac{500}{\$10}$ (or 50) is greater than $\frac{1000}{\$100}$ (or 10). The marginal product per dollar of labor exceeds the marginal product per dollar of capital. So, the firm should use more labor and less capital to produce the 50 widgets.

36. In the long run, assume a firm uses both labor and capital to produce 25 units of output. The marginal product of the last unit of labor being employed is 100; the marginal product of the last unit of capital being employed is 500. The wage rate of labor is $\$10$. If the firm is minimizing the cost of producing 25 units of output, what must be the unit price of capital?

(A) $\$5$

(B) $\$10$

(C) $\$25$

(D) $\$50$ ✓

(E) $\$500$

Answer D

Correct. The rule for minimizing cost in the long run is to equate the marginal product per dollar of each input, i.e.,$\frac{\text{MP}_\text{L}}{\text{P}_\text{L}} = \frac{\text{MP}_\text{K}}{\text{P}_\text{K}}$. Thus, $\frac{100}{\$10} = \frac{500}{\text{P}_\text{K}}$. The price of capital must be $\$50$ per unit to minimize the cost of producing the 25 units of output.

37. A firm is currently producing 3,000 units of output daily by employing 20 units of labor at a price of $100 per unit and 40 units of capital at a price of $40 per unit. The marginal product of the last unit of labor employed is 50, and the marginal product of the last unit of capital employed is 30. In order to minimize its production costs, the firm should do which of the following?

5.3 perfect competitive factor market

(A) Employ more labor and less capital because the marginal product of labor is greater than the marginal product of capital.

(B) Employ less labor and more capital because the firm is currently spending $2,000 on labor and only $1,600 on capital.

(C) Employ more labor and less capital because the firm already employs 40 units of capital and only 20 units of labor.

(D) Employ less labor and more capital because the marginal product per dollar spent on labor is less than the marginal product per dollar spent on capital. ✓

(E) Employ less labor and more capital because a unit of labor costs $100 while a unit of capital costs only $40.

Answer D

Correct. The cost-minimizing combination of inputs is based on marginal product per dollar spent. In this example, the marginal product per dollar spent on labor is 0.5 $(50/\$100)$, which is less than the marginal product per dollar spent on capital, which is 0.75 $(30/\$40)$. Thus, the firm can minimize the cost of the given output by using less labor and more capital.

38.

	Capital	Labor
Marginal Product	25 units	5 units
Hourly Price	$200	$50

Assume that capital and labor are the only two inputs used by a clinic. The table shows the hourly marginal product and the hourly price of capital and labor at the current output level. To minimize costs at the current output level, the clinic should change the hiring of capital and labor in which of the following ways?

(A) The amount of capital hired should decrease, and the amount of labor hired should increase.

(B) The amount of capital hired should decrease, and the amount of labor hired should decrease.

(C) The amount of capital hired should increase, and the amount of labor hired should decrease. ✓

(D) The amount of capital hired should increase, and the amount of labor hired should increase.

(E) There should be no change to the quantity of capital or labor hired.

Answer C

Correct. To minimize costs, a firm must equate the marginal product of labor per dollar ($\mathrm{MP_L}/\mathrm{P_L}$) with the marginal product of capital per dollar ($\mathrm{MP_K}/\mathrm{P_K}$). Since $\mathrm{MP_K}/\mathrm{P_K}$ is $25/\$200 = 0.125/\$$ is greater than the $\mathrm{MP_L}/\mathrm{P_L} = 5/\$50 = 0.10/\$$ in this case, the firm should employ more capital

5.3 perfect competitive factor market

and less labor in the production process to minimize its costs.

39. Mary manages a manufacturing plant. Two inputs are used to produce the product. To minimize the costs of production, Mary should continue to hire inputs up to the level at which the

(A) price of each input is equal

(B) marginal product of each input is equal

(C) (marginal product $\times$ price) of each input is equal

(D) (marginal product $\div$ price) of each input is equal ✓

(E) (marginal product $-$ price) of each input is equal

Answer D

Correct. The cost-minimizing combination of inputs is realized when the marginal product per dollar spent on each input is equal. As $\frac{\text{MP}}{\text{Input Price}}$ is equal for each input then this combination of inputs will produce the output level at the lowest total cost.

40. Pickleco, a pickle-producing firm, hires labor and capital in perfectly competitive factor markets. The firm is minimizing its costs at the current production level. The marginal product of labor is 100 units and the marginal product of capital is 60 units. If the rental price of capital is $\$12$, what is the wage Pickleco is paying its workers?

(A) $\$5.00$

(B) $\$8.33$

(C) $\$12.00$

(D) $\$20.00$ ✓

(E) $\$40.00$

Answer D

Correct. The cost-minimizing combination of inputs is realized when the marginal product per dollar spent of each input is equal, that is $\frac{\text{MP}_\text{k}}{\text{P}_\text{k}} = \frac{\text{MP}_\text{l}}{\text{W}}$. This least-cost hiring rule $\left(\frac{60}{\$12} = \frac{100}{\text{W}}\right)$ can be used to calculate the wage that Pickelco will pay to its workers as $\$20$.

41. At a given quantity of output of a firm, if the marginal product of capital equals the marginal product of labor and the wage rate equals the rental rate of capital, then

5.3 perfect competitive factor market

(A) the firm is producing that quantity at the lowest cost ✓
(B) the firm should use more capital and less labor to minimize cost
(C) the firm should use less capital and more labor to minimize cost
(D) the firm should use more of both inputs to increase quantity
(E) the firm should use less of both inputs to increase quantity

Answer A

Correct. The cost-minimizing combination of inputs is realized when the marginal product per dollar spent on each input is equal. The firm is producing the quantity at the lowest cost because the marginal product of capital per the rental rate of capital equals the marginal product of labor per the wage rate.

42. Marginal revenue product is defined as the

(A) change in income that occurs when an individual works additional hours
(B) change in total revenue that occurs when one additional unit of the good is produced
(C) change in total revenue that occurs when one additional unit of an input is employed ✓
(D) total revenue divided by the quantity of labor employed
(E) change in total cost that occurs when one additional unit of an input is employed

43. The demand curve for labor for a monopolist that faces a perfectly competitive factor market is called the

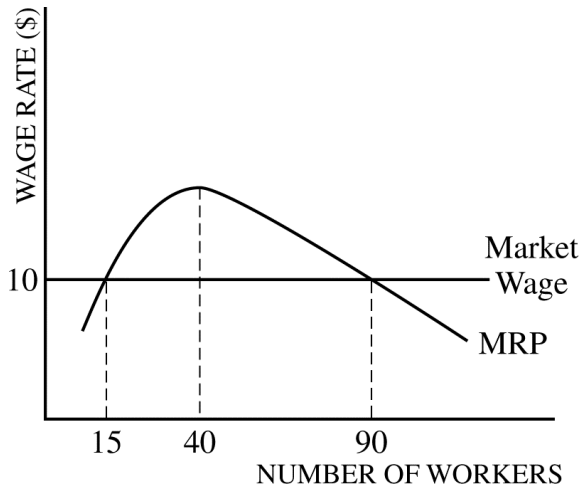
(A) average product curve
(B) marginal product curve
(C) marginal revenue curve
(D) marginal revenue product curve ✓
(E) value of the average product curve

44. The last worker currently employed by a firm has a marginal product of 3 units per hour and is paid $20 per hour. Assuming that both the labor market and product market are perfectly competitive and that the product's price is $5 per unit, the firm should do which of the following?

(A) Decrease wages
(B) Increase wages
(C) Employ more workers
(D) Employ fewer workers ✓
(E) Increase the product's price

5.3 perfect competitive factor market

45.



The graph above shows the marginal revenue product (MRP) and the market wage rate for a profit-maximizing firm. Which of the following is true of the firm's hiring of labor?

(A) It should hire 15 workers.

(B) It should hire between 15 and 40 workers.

(C) It should hire 40 workers.

(D) It should hire between 40 and 90 workers.

(E) It should hire 90 workers. ✓

A perfectly competitive firm currently produces $1{,}000$ units of output and hires its resources in a perfectly competitive factor market. It uses both labor and capital as inputs. The price of labor is $\$40$; the price of capital is $\$100$. The marginal product of labor is 8 units, and the marginal product of capital is 10 units.

46. Which of the following must be true?

(A) The firm is currently maximizing its profit.

(B) The firm can produce more than $1{,}000$ units without increasing the total cost if it uses more labor and less capital. ✓

(C) The firm can produce more than $1{,}000$ units without increasing the total cost if it uses more capital and less labor.

(D) The firm can reduce the cost of producing $1{,}000$ units by using less capital and employing the same amount of labor.

(E) The firm can reduce the cost of producing $1{,}000$ units by employing less labor and using the same amount of capital.

Answer B

Correct. To minimize its costs, a firm should use the combination of labor and capital such that the

5.3 perfect competitive factor market

marginal product of the last dollar spent on labor is equal to the marginal product of the last dollar spent on capital. From the given data, the marginal product of the last dollar spent on labor $(8/\$40)$ is greater than the marginal product of the last dollar spent on capital $(10/\$100)$. Therefore, the firm can produce more than $1{,}000$ units of output by using more labor and less capital.

The following two questions refer to the data in the table below.

A perfectly competitive firm operates with a fixed amount of capital that costs $\$1{,}000$ per day. Labor is the only variable input. The firm hires labor in a perfectly competitive labor market at $\$100$ per day per worker. The table below shows the firm's production function.

Number of Workers Hired	Quantity of Output (units)
0	0
1	10
2	30
3	54
4	75
5	85
6	90

47. The firm will maximize profit in the short run if it

(A) shuts down and the price is $\$10$

(B) produces 30 units and the price is $\$4$

(C) produces 30 units and the price is $\$10$

(D) produces 54 units and the price is $\$4$

(E) produces 85 units and the price is $\$10$ ✓

Answer E

Correct. If the firm produces 85 units and sells at $\$10$, the marginal product of the fifth worker is 10 units. The value of the marginal product of labor (marginal revenue product) equals the marginal factor

5.3 perfect competitive factor market

cost. That is, $(\$10 \times 10) = \$100$, which is equal to the marginal factor cost of $\$100$. Thus, profit is maximized.

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48. Which of the following will occur in a perfectly competitive labor market if Firm X's demand for labor decreases?

(A)

Equilibrium Market Wage Rate	Equilibrium Market Wage Rate
Increase	Increase

(B)

Equilibrium Market Wage Rate	**Employment by Firm X**
No change	No change

(C) ✓

Equilibrium Market Wage Rate	Employment by Firm X
No change	Decrease

(D)

Equilibrium Market Wage Rate	**Employment by Firm X**
Decrease	Decrease

(E)

Equilibrium Market Wage Rate	Employment by Firm X
Decrease	No change

5.3 perfect competitive factor market

49. Let W denotes nominal wage, P the output price, and MP_L the marginal product of labor. Which of the following relationships correctly estimates the marginal cost (MC) of production for a perfectly competitive firm in the short run?

(A) $MC = P/MP_L$

(B) $MC = P \times MP_L$

(C) $MC = W \times MP_L$

(D) $MC = W/MP_L$ ✓

(E) $MC = MP_L/W$

50.

Quantity of Labor	Marginal Product	Marginal Revenue Product
0	---	---
1	20	$40
2	14	28
3	12	24
4	8	16
5	6	12
6	4	8

Based on the table for a perfectly competitive firm above, if the wage rate for labor is $15, how many units of labor should the firm hire?

(A) 2

(B) 3

(C) 4 ✓

(D) 5

(E) 6

5.3 perfect competitive factor market

Question is based on the chart below for a firm that is perfectly competitive in both the labor and product markets, showing how much daily output a firm can produce using different numbers of workers.

Number of Workers	Output
1	3
2	9
3	16
4	21
5	23
6	24

51. If output sells for $20 per unit and the daily wage is $100 per worker, how many workers should the firm hire to maximize profit?

(A) 1
(B) 2
(C) 4 ✓
(D) 5
(E) 6

52.

Number of Workers	Parts per Week
0	0
1	100
2	190
3	270
4	340
5	400
6	450
7	480

The table above describes the production function of an auto parts manufacturer. Assume that the firm can hire as many workers as it wants at the market wage rate of $600 per week per worker and sell as many auto parts as it wants at the price of $10 per part. To maximize profits, the firm should hire

5.3 perfect competitive factor market

(A) 0 workers
(B) 1 worker
(C) 3 workers
(D) 5 workers ✓
(E) 7 workers

53. Which of the following indicates that a perfectly competitive firm has hired the profit-maximizing amount of labor?

(A) The total product of labor exceeds the total real wage payments to workers.
(B) The average product of labor exceeds the real wage paid to workers.
(C) The marginal revenue product of labor is below the wage paid to workers.
(D) The marginal revenue product of labor is above the wage paid to workers.
(E) The marginal revenue product of labor equals the wage paid to workers. ✓

Answer E

Correct. When the marginal revenue product of labor is equal to the marginal factor cost (which is the same as the market wage), the firm is using the profit-maximizing quantity of labor, and profit is maximized.

54. If a competitive firm pays its workers the value of the marginal product of the last worker hired, which of the following is true?

(A) The firm will not earn any economic profits.
(B) Workers will look for employment elsewhere.
(C) The wage will be less than the marginal product.
(D) The firm will not maximize profits.
(E) The contribution of the last worker hired to the firm's profit will be zero. ✓

55. If a firm employs only labor and capital in its production process, which of the following best describes the optimal combination of inputs for the firm in the long run?

(A) The marginal product per dollar spent on labor is equal to the marginal product per dollar spent on capital. ✓
(B) The marginal product of labor is equal to the marginal product of capital.
(C) The total product of labor is equal to the total product of capital.
(D) The marginal product of labor and capital are both zero.
(E) All marginal products are equal to all average products.

5.3 perfect competitive factor market

The following questions refer to the diagram below, which shows the short-run production function of a perfectly competitive firm that produces potatoes using one variable input, labor.

Number of Workers	Potatoes Harvested (pounds per hour)
0	0
1	3
2	7
3	10
4	12
5	13
6	12

56. If the firm can sell as many potatoes as it wants for $2 per pound and has to pay each worker $5 per hour, how many workers should the firm employ to maximize profits?

(A) 1
(B) 2
(C) 3 ✓
(D) 4
(E) 5

57.

- The marginal product of labor equals 250 units of output.
- The marginal product of capital equals 750 units of output.
- The price of labor is $50 per person.
- The price of capital is $100 per unit.

Given the information above, which of the following is true for a firm buying its labor and capital inputs in a perfectly competitive market?

5.3 perfect competitive factor market

(A) The firm is producing its current level of output with the least-cost combination of labor and capital.

(B) The firm is maximizing profits with its current combination of inputs.

(C) The firm's level of output will remain the same if 1 unit of capital is substituted for 2 units of labor.

(D) The firm's input costs will decrease if 2 units of labor are substituted for 1 unit of capital.

(E) The firm can reduce the cost of its current level of output by laying off workers and employing more capital. ✓

58. The wage rate is $10 per hour and the last worker hired by the firm increased output by 100 units. Computers rent for $100 per hour and the last computer rented by the firm increased output by 2,000 units. To minimize costs the firm should

(A) hire more workers and rent more computers because the marginal revenue products of both workers and computers are greater than their respective prices.

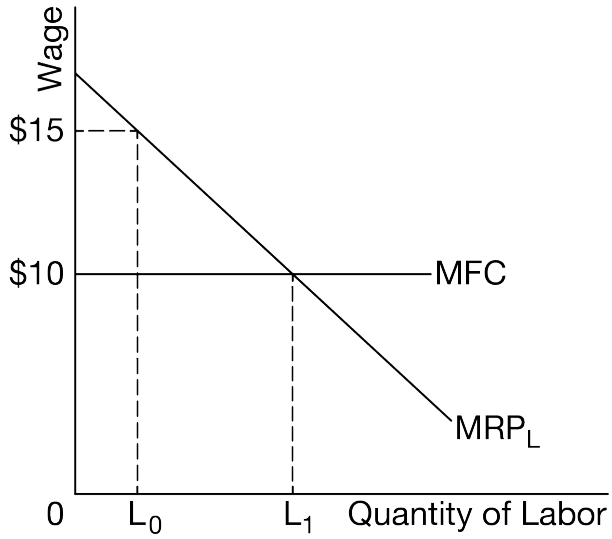
(B) hire more workers and reduce the number of computers rented because workers are cheaper than computers

(C) lay off workers and rent more computers because computers produce more output per dollar of additional expenditure ✓

(D) lay off workers and rent more computers because computers produce more output

(E) keep the same number of workers and computers because the marginal revenue products of both workers and computers are positive

59.



Based on the graph above, which of the following statements is true?

(A) Hiring unit of labor L_0 will add $15 to a firm's economic profits.

(B) The market wage rate is $10 per unit of labor. ✓

(C) The profit-maximizing firm will hire its labor at less than $10 per unit.

(D) From 0 units of labor to L_0 units of labor, there are increasing marginal returns to labor.

(E) Along the firm's marginal revenue product of labor curve, the price of output falls as more labor is hired and more output is produced.

5.3 perfect competitive factor market

Answer B

Correct. The profit-maximizing competitive firm is a price taker, paying $\$10$ per unit of labor. It will hire labor until the equality of the marginal revenue product of labor and the marginal factor cost of labor, L_1, on this graph.

The table shows the short-run production of a firm that produces and sells its product in a perfectly competitive market.

Number of Workers	Quantity of Output
0	0
1	8
2	15
3	21
4	26
5	30

60. If the firm sells its product at the market price of $10 per unit, the marginal revenue product of the fourth worker is

(A) $40

(B) $50 ✓

(C) $65

(D) $260

(E) $300

Answer B

Correct. The marginal revenue product of the fourth worker is equal to the marginal product of the fourth worker (that is, the amount that production increases when a fourth worker is added) multiplied by the price of the product. In this case, $\text{MRP} = (26 - 21) \times \$10 = \$50$.

61. If the firm sells its product at the market price of $10 per unit, how many workers should the firm employ to maximize profit if the wage rate is $55?

5.3 perfect competitive factor market

(A) One
(B) Two
(C) Three ✓
(D) Four
(E) Five

Answer C

Correct. The third worker's marginal revenue product is $6 \times \$10 = \$60$, which is greater than the wage, \$55. However, the fourth worker's marginal revenue product is $5 \times \$10 = \$50$, which is less than the wage, \$55. Thus, the firm will maximize profit by employing three workers.

Hours of Labor	Marginal Product
0	0
1	10
2	12
3	9
4	7
5	4

The table above shows how a firm's hourly level of output changes as more of the labor input is employed. The firm sells its output and hires labor in perfectly competitive markets. The wage paid to labor is $\$10$ per hour, and the price of the firm's output is $\$2$ per unit.

62. Based on the data in the table, the marginal revenue product of the fourth hour of labor is equal to

(A) $\$4$
(B) $\$7$
(C) $\$14$ ✓
(D) $\$26$
(E) $\$70$

5.3 perfect competitive factor market

Answer C

Correct. The marginal revenue product of labor (MRPL) is defined as the marginal product of labor (MPL) times marginal revenue (MR). The MRPL of the fourth hour of labor is equal to $7 \times \$2 = \$14$.

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63. For a firm operating in perfectly competitive product and factor markets, the marginal revenue product of labor is equal to the

(A) marginal product of labor

(B) price of labor multiplied by the marginal product of labor

(C) price of labor multiplied by the price of the output

(D) price of the output divided by the price of labor

(E) price of the output multiplied by the marginal product of labor ✓

Answer E

Correct. The marginal revenue product of labor refers to the additional revenue received by the firm from hiring additional quantities of labor. Since price is equal to marginal revenue for firms selling products in perfectly competitive markets, the marginal revenue product of labor for such firms operating in perfectly competitive factor markets is equal to the output price multiplied by the marginal product of labor.

64. Max Company produces electronic gadgets using labor and capital. If Max increases labor and decreases capital to produce the same quantity of gadgets, what will happen to the marginal product (MP) of labor and the MP of capital?

5.3 perfect competitive factor market

(A) ✓

MP of Labor	MP of Capital
Decrease	Increase

(B)

MP of Labor	MP of Capital
Decrease	No change

(C)

MP of Labor	MP of Capital
Increase	Increase

(D)

MP of Labor	MP of Capital
Increase	Decrease

(E)

MP of Labor	MP of Capital
No change	Increase

Max employs both labor and capital to produce toy trains. Currently the last unit of labor employed has a marginal product of 15 units. The last unit of capital employed has a marginal product of 40 units. The price of labor is $3 per unit, and the price of capital is $10 per unit.

65. Which of the following employment decisions should Max follow to use the least-cost combination of labor and capital to produce the current quantity of toy trains?

5.3 perfect competitive factor market

(A)

Employment of Labor	Employment of Capital
Increase	No change

(B)

Employment of Labor	Employment of Capital
Decrease	Increase

(C)

Employment of Labor	Employment of Capital
Decrease	Decrease

(D)

Employment of Labor	Employment of Capital
Increase	Increase

(E) ✓

Employment of Labor	Employment of Capital
Increase	Decrease

5.3 perfect competitive factor market

The following question is based on the table provided, showing hourly production data and total revenue for a perfectly competitive firm.

Number of Workers Hired	Total Revenue ($)	Quantity Produced
0	0	0
1	50	10
2	95	19
3	135	27
4	170	34
5	200	40

66. What is the marginal revenue product of the fourth worker?

(A) $35 ✓
(B) $34
(C) $30
(D) $7
(E) $5

Answer A

Correct. The marginal revenue product of labor is calculated as the change in total revenue/the change in number of workers hired. As the number of workers hired increases from 3 workers to 4 workers, total revenue increases from $135 to $170; thus, the marginal revenue product of the fourth worker is $35.

5.3 perfect competitive factor market

67. Robots and workers are substitutes in production and are both variable inputs. Both robots and workers are hired in perfectly competitive factor markets. If the daily rental price of robots increases, which of the following will most likely happen to wages paid to workers and the number of workers employed in the short run?

	Wages	Number of Workers Employed
A	Decrease	Decrease
B	Decrease	Increase
C	Increase	Decrease
D	Increase	Increase
E	No change	No change

(A) A
(B) B
(C) C
(D) D ✓
(E) E

Answer D

Correct. If the factors of production are variable, firms will substitute for the relatively cheaper factor if the price of one of the factors increases. In this case, the price of robots increased, incentivizing the firm to increase its demand for workers in the short run. This increase in demand for workers will increase the wage rate and increase the number of workers employed in the short run.

68.

Number of Workers	Quantity of T-Shirts
0	0
1	24
2	44
3	60
4	72
5	80

The table provided shows the daily output for a profit-maximizing T-shirt firm that is operating in a perfectly competitive product market and hires labor from a perfectly competitive labor market. If each T-shirt sells for $10 and the equilibrium daily wage is $100, how many workers will the firm hire to maximize profit?

5.3 perfect competitive factor market

(A) 1
(B) 2
(C) 3
(D) 4 ✓
(E) 5

Answer D

Correct. Firms operating in perfectly competitive product and labor markets will maximize profit by hiring workers as long as the marginal revenue product of the last worker hired (marginal product of labor multiplied by the price of the good) is greater than or equal to the wage rate. When the firm hires four workers, the marginal product of labor is 12 units ($=\frac{72-60}{4-3}$) and the price of the T-shirt is \$10, so the marginal revenue product of labor is \$120 which is greater than the wage rate; thus, the profit-maximizing number of workers to hire is 4 workers. Hiring more than four workers will result in a wage rate exceeding the marginal revenue product, thus not profit-maximizing.

ZBA is a typical profit-maximizing firm in a perfectly competitive market that produces and sells widgets at \$5 each. ZBA hires labor in a perfectly competitive labor market. The table shows ZBA's production function for widgets.

Number of Workers	Total Output/Hour
1	3
2	7
3	12
4	14
5	15

69. What is the marginal revenue product of the third worker?

(A) \$5
(B) \$12
(C) \$25 ✓
(D) \$36
(E) \$60

5.3 perfect competitive factor market

Answer C

Correct. The marginal revenue product of labor is calculated as the change in total revenue/the change in number of workers hired or the product of the marginal physical product and the marginal revenue. As the number of workers hired increases from 2 workers to 3 workers, total revenue increases by $25, marginal physical product of 5 multiplied by the marginal revenue of $5.

70. If the hourly wage is $9, how many workers will the firm hire to maximize profit?

(A) 1
(B) 2
(C) 3
(D) 4 ✓
(E) 5

Answer D

Correct. A firm operating in perfectly competitive output and labor markets will maximize profit by hiring workers as long as the marginal revenue product of the last worker hired (marginal product of labor multiplied by the price of the good) is greater than or equal to the wage rate. In this example, the marginal revenue product of the 4th worker is $10 (marginal physical product of 2 units times marginal revenue of $5) and is greater than the wage rate ($9). Since the marginal revenue product is less than the wage for the fifth worker, the firm should only employ four workers to maximize profit.

CDDAC
DDEBE
CDCCD
EEACE
CBBCC
EAEAD
DCD